METHODOLOGY DOCUMENT

Real-Time Athlete Performance Analytics and Anomaly Detection

1. Data schema

Required fields: timestamp, athlete_id, heart_rate, speed, acceleration_x, acceleration_y, acceleration_z. Optional supervised target: performance_score. Record units and sensor sampling frequency. Validate data provenance and athlete consent.

2. Preprocessing

Parse timestamps, coerce numeric sensor columns, remove invalid records, sort by athlete and timestamp, and resolve duplicate athlete/timestamp records. Document missing data and any filtering. Do not silently mix sensor units or devices.

3. Feature engineering

Acceleration magnitude = sqrt($ax^2 + ay^2 + az^2$). Calculate shifted rolling heart-rate mean/std per athlete, delta heart rate, delta speed, acceleration-magnitude change, normalized sensor features, and a documented exertion index. Shift rolling windows to prevent current/future target leakage.

4. Performance prediction

Train RandomForestRegressor on engineered features to estimate performance_score. Define the target independently and explain its measurement. GroupShuffleSplit by athlete is used in the demo to reduce identity leakage. Report MAE, RMSE, R^2 and split details. The synthetic target is only a pipeline example.

5. Anomaly detection

Fit IsolationForest on selected sensor-derived features. The estimator labels inliers as 1 and outliers as -1. Set contamination as an initial assumption, then calibrate using expert review and labeled data if available. Statistical anomaly does not imply injury or diagnosis.

6. Alert logic

Generate alerts from configurable heart-rate and acceleration thresholds plus the anomaly flag. Record timestamp, athlete identifier, triggered conditions, and severity. Thresholds must be specific to device units, sport, context, and expert-reviewed protocols.

7. Evaluation and reporting

Regression: MAE, RMSE, R^2. Anomaly detection with labels: precision, recall, F1, confusion matrix. Without labels: anomaly rate and human-reviewed examples. Streaming system: latency, throughput, data loss, alert delay. Clearly distinguish synthetic demo results from real-data evaluation.

8. Reproducibility and ethics

Record random seed, package versions, hardware, sensor details, data version, target definition, split method, and thresholds. Protect athlete privacy, obtain consent, restrict access, and avoid public uploads of identifiable sensor records. Do not use this prototype for clinical decisions or return-to-play clearance.